

PRE-TRAINING ASSIGNMENT

CNC Motion & Control Research

Connect industrial automation principles to CNC machinery through the design of a simple two-axis pen plotter.

ESTIMATED EFFORT	OUTPUT	BRING
60-90 minutes	Concise notes and completed tables	Research findings and source links

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1. Purpose and Objective

The purpose of this assignment is to connect industrial automation principles to real CNC systems. By researching motion mechanisms, motors, feedback devices, drives, and spindle control, you will understand how automation components interact within a CNC machine. You will then apply these principles by proposing a safe, practical two-axis pen plotter.

Your objective: Explain each concept clearly in your own words and support every design decision with sound technical reasoning. The strength of your justification matters more than the price or complexity of the components you select.

Expected Learning Outcomes

- Describe how industrial automation components and signal flow apply to CNC machines.
- Compare common CNC axis-motion mechanisms.
- Identify major motor types and distinguish between stepper and servo systems.
- Explain feedback, encoders, motor drivers, STEP/DIR commands, and VFDs.
- Propose a complete, practical architecture for a two-axis pen plotter.

How to Complete the Task

- Use reliable sources, including manufacturer documentation, textbooks, and technical training materials.
- List at least three source links at the end of this document.
- Keep your answers concise. Use keywords, short explanations, and small sketches where appropriate.
- Be prepared to justify one design decision during the training discussion.

2. Research Task A - CNC Axis Motion

Electric motors produce rotary motion, whereas CNC axes generally require linear motion. Research how each method transmits or converts motor rotation into linear axis movement.

A1. Method Overview

Method	How it works	Main advantages	Main limitations	Typical applications
Timing belt	Motor turns a pulley; a toothed belt loops between it and an idler pulley, with the carriage clamped to the belt. Teeth stop slip, so belt travel = carriage travel.	Low cost, low inertia, fast, quiet, simple to install	Belt stretch/elasticity adds compliance and some backlash; lower stiffness; limited load	3D printers, laser cutters, plotters, light-duty routers
Lead screw	A threaded rod is turned by the motor; a nut fixed to the carriage rides the thread, sliding along the screw as it rotates.	Cheap, self-locking (holds position unpowered), compact	Sliding friction = wear, heat, low efficiency (~30-50%); backlash unless preloaded	3D printers, low-cost stages, Z-axes (benefit from self-locking)
Ball screw	Same rotary-to-linear idea as a lead screw, but recirculating balls roll between screw and nut instead of sliding.	High efficiency (>90%), high precision, low/near-zero backlash when preloaded, long life	Costly; not self-locking (needs brake on vertical axes); screw whip limits length/speed	Precision mills, lathes, machining centers
Rack and pinion	A motor (usually through a gearbox) drives a pinion gear that walks along a straight or helical rack fixed to the axis.	Unlimited travel (racks bolt together end to end), high speed, high stiffness, tolerant of dust	Backlash unless dual-motor/spring preloaded; less accurate than ball screw; gear noise	Large-format routers, plasma/laser gantries, long-travel X axes

A2. Comparison Matrix

Use relative ratings such as *Low*, *Medium*, or *High*, and add a brief note when clarification is needed.

Method	Speed	Accuracy	Backlash	Cost	Load capacity
Timing belt	High	Medium	Medium (belt stretch)	Low	Low
Lead screw	Low	Medium	Medium-High (unless preloaded)	Low	Medium
Ball screw	Medium-High	High	Low (preloaded)	High	High
Rack and pinion	High	Medium	Medium (unless dual-pinion)	Medium	High

A3. Engineering Choice

1. Which method would you select for a small pen plotter, and why?

Response: Timing belt (GT2, 2 mm pitch). A pen only needs enough force to overcome its own drag and carriage inertia, the belt's lower stiffness and slight compliance never show up as visible error in a drawing.

Low mass and low cost also matter more here than in cutting: a light belt-driven carriage lets a small stepper accelerate and decelerate quickly through curves, which is what gives a plotter smooth, round strokes instead of faceted ones.

What changes if the same frame is later used for cutting instead of drawing?

Response: Cutting forces are orders of magnitude higher than pen drag, so belt stretch and backlash that were invisible while drawing become real dimensional error and can even skip belt teeth under load.

At minimum the transmission would need to move to a ball screw or a preloaded/dual-pinion rack-and-pinion for stiffness; motors and drivers would need re-sizing for the higher torque, and a spindle plus VFD (see section C4) would have to be added.

3. Research Task B - Motors for CNC

B1. Five-Minute Motor Scan

Identify the following motor types and write one sentence explaining how each operates or where it is typically used: brushed DC, brushless DC (BLDC), AC induction, stepper, and servo systems.

Motor family	Basic operating idea	One common application
Brushed DC	Brushes and a commutator switch current through rotor windings as it spins, so the rotor is always pulled toward the next pole of the fixed magnet field.	Cordless hand tools, small pumps/fans
BLDC	A permanent-magnet rotor is spun by stator coils switched electronically (Hall-sensor or sensorless back-EMF), so no brushes are needed.	Drone motors, spindle motors, high-speed cooling fans
AC induction	A rotating magnetic field from multi-phase AC in the stator induces current in the rotor bars; the rotor's induced field chases the stator field, producing torque.	CNC spindle motors (via VFD), industrial pumps and conveyors
Stepper	Stator coil pairs are energized in sequence; a toothed rotor snaps to align with each energized pole, producing motion in discrete angular steps.	3D printers, small CNC routers/plotters
Servo system	A motor (often BLDC/PMSM) is paired with an encoder in a closed loop; the drive compares commanded vs. actual position/speed and continuously corrects the current.	Machining-center axes, robot-arm joints

B2. Detailed Comparison - Stepper vs Servo

Complete the comparison by considering the full system: motor, driver, feedback device, controller, and tuning requirements.

Criterion	Stepper system	Servo system
Control / feedback	Open-loop — no position feedback; the drive just counts the pulses it is sent	Closed-loop — encoder continuously reports actual position/velocity back to the drive

Criterion	Stepper system	Servo system
Torque at low speed	High; near-maximum holding torque at zero/low speed	Moderate at low speed, rated torque available on demand
Torque at high speed	Drops sharply above a few hundred RPM (back-EMF limits current)	Stays comparatively flat across a wide speed range
Holding at standstill	Strong passive holding torque from the energized coils, even without active control	No inherent holding torque; the control loop must actively hold position (a brake is added for power-off holding)
Accuracy and repeatability	Good, set by step angle / microstepping — as long as no steps are missed	Very good, set by encoder resolution; self-correcting even under load changes
Missed-step / position-error behavior	Can silently lose sync if overloaded — no way for the controller to detect it	A following-error alarm trips if the axis can't keep up, so faults are caught, not hidden
Efficiency, heat, and noise	Windings stay energized even at standstill, so it runs warm and can resonate/whine at low speed	Current tracks actual load, so it runs cooler and generally quieter
Setup and tuning	Simple — set current and microstep setting, then run	Needs loop (PID) tuning for stability; more commissioning effort
Cost and complexity	Low cost, simple driver, two wires of feedback not required	Higher cost — motor + encoder + drive — and more complex wiring
Typical CNC use	Small routers, plotters, 3D printers, hobby-class machines	Production machining centers, high-speed or heavily loaded axes

B3. Selection Question

Which motor system would be the most suitable starting point for a lightweight pen plotter? Give two reasons and identify one limitation.

Response: Stepper system.

Reason 1: pen-drag loads are tiny, so the stepper's low-speed torque advantage covers the job with no real risk of missed steps.

Reason 2: it is far simpler and cheaper to drive directly from the RD controller's STEP/DIR outputs, and it matches the two stepper motors already supplied for this build.

Limitation: no feedback — a jam, a snagged cable, or too-aggressive acceleration can cause missed steps that the controller cannot detect, so the drawing comes out shifted with no fault warning.

4. Research Task C - Control, Feedback, and Drives

C1. Open-Loop vs Closed-Loop Control

- Define open-loop control and give one CNC example.
- Define closed-loop control and identify the information returned to the controller or drive.
- Describe what may happen if an open-loop stepper motor is overloaded.
- Explain how feedback enables a system to detect or correct a position error.

Your answer / notes:

Open-loop

the controller sends a command with no confirmation the motion actually happened. CNC example: a stepper axis running purely on counted STEP pulses, trusting the motor followed every one.

Closed-loop

the controller/drive receives feedback — typically position, and often velocity, from an encoder — and compares it to the command. Information returned to the drive: actual shaft/axis position (and derived velocity).

Open-loop stepper overloaded

the rotor falls out of sync with the commanded phase sequence and "stalls" or skips steps. The controller keeps counting pulses as if the axis moved, so a permanent position offset builds up unnoticed until it shows up on the part or drawing.

Feedback and error correction

encoder counts are compared to the commanded position continuously; any difference (following error) becomes a correction signal, and the drive adjusts current/velocity to drive that error toward zero before it can accumulate.

C2. Encoder Role and Types

Explain how an encoder converts mechanical motion into electrical feedback. Then compare the two encoder types below.

An encoder attaches to a moving part (a motor shaft, a linear slide, etc.) and generates electrical pulses or signals as that part moves. By counting or decoding those signals, a controller can determine position, speed, and direction. It turns motion into electrical feedback in four basic steps:

1. **Sensing** – A disk or strip with alternating patterns (optical, magnetic, or capacitive) is attached to the moving part.
2. **Signal generation** – As it moves, a sensor (light detector, Hall sensor, etc.) detects each pattern transition and outputs an electrical pulse.
3. **Counting** – A controller counts these pulses to calculate position and speed (pulse frequency = speed).
4. **Direction** – Two sensors offset by 90° (quadrature) let the controller tell which way it's moving by comparing which pulse leads the other.

Encoder type	What it reports	What happens after power is removed	Advantages	Limitations / uses
Incremental	Relative pulses per increment of motion (count + direction)	Reference is lost, position is unknown until the axis is re-homed	Simple, inexpensive, high resolution for the price	Needs a homing routine every power-up; can drift if pulses are missed
Absolute	A unique digital code representing the actual shaft position right now	Position is retained (single-turn, or full turn-count on multi-turn versions)	No homing needed; knows its position immediately at power-up	Higher cost; more complex interface (e.g., BiSS/SSI); often unnecessary on simple axes

Encoder role summary: *an encoder converts the mechanical rotation of a shaft into an electrical signal — quadrature pulses or a digital code — that the drive/controller can read, closing the loop between commanded and actual motion so any error can be measured and corrected in real time.*

C3. Motor Driver and STEP/DIR

- State the purpose of a motor driver and explain why a controller should not supply power directly to a motor.
- Explain the STEP signal: what motion does each pulse normally command?
- Explain the DIR signal: what does its logic level determine?
- Describe how microstepping and pulse-per-unit settings affect commanded travel.
- Review the documentation for the RD controller and selected driver to confirm signal type, voltage levels, grounding requirements, and compatible wiring.

Your answer / notes:

Purpose of a motor driver

A motor driver takes the small, weak signals from the controller and turns them into the strong electrical power a motor actually needs to move. The controller only produces logic-level signals, usually around 3.3 to 5 volts and very little current. A motor needs much more current and, in the case of a stepper motor, its coils need to be switched on and off in a very specific pattern to spin properly. The driver handles all of that. If you connected the controller straight to the motor, there wouldn't be enough power to move it, and the weak signal could actually damage the controller. So the driver acts as the middleman: it listens to the controller and supplies the motor with properly timed, high current power.

STEP signal

The STEP wire tells the driver when to move. Every time this signal pulses (goes from low to high, or high to low, depending on the driver), the driver moves the motor by one small fixed amount, called one step or one microstep. So the STEP signal doesn't control speed or direction by itself. It just says "move now." Sending pulses faster makes the motor spin faster, and sending more pulses makes it travel farther.

DIR signal

The DIR wire tells the driver which way to move. It only has two states, high or low. One state means turn clockwise, the other means turn counter-clockwise. The controller sets this wire before sending step pulses, and the driver keeps moving in that direction until DIR changes.

Microstepping and pulses-per-unit

A motor normally has a fixed number of full steps per rotation (for example, 200). Microstepping is a feature of the driver that breaks each of those full steps into smaller pieces by carefully controlling the current in the coils. This lets the motor move in smaller, smoother increments instead of big jumps, though each microstep produces slightly less torque than a full step.

Because of this, the controller needs to know exactly how many pulses correspond to one unit of real world movement, like one millimeter. This is called the pulses-per-unit or steps/mm setting. It's calculated from the motor's steps per revolution, multiplied by the microstepping setting, divided by how far the mechanism actually moves in one revolution. If this number is set wrong, every distance the machine tries to move will come out wrong, even though the motor is working correctly.

Checking documentation: brief of findings

RDC6445S(EC) controller

The pulse and direction outputs are open-collector (OC), optically isolated, and the manual specifies the pulse and directional signals are both OC outputs, and the controller must be common anode with the motor driver. In practice this means the controller sinks current rather than sourcing voltage, so the driver's PUL+/DIR+ inputs get tied to a positive supply (often the controller's own 5V output pin) while PUL-/DIR- connect to the controller's OC output pins. The pulse edge polarity and direction polarity are both configurable in the controller's software, which is useful if a motor ends up moving the wrong way. [Aliyuncs](#)

3DM580 driver for the 573S15-L motor

This is a 3-phase driver with 3 optically isolated control inputs — PUL, DIR, and ENA — that accept differential and single-ended signals (open-collector and PNP outputs). Voltage-wise, logic current is 7-16mA (typically 10mA), and HIGH is 3.5-5V, LOW is 0-0.5V. If you feed it 12V or 24V instead of 5V, the manual says to series-connect a current-limiting resistor — don't wire higher voltages straight in. Power input is 20-50VDC, and microstepping is set via DIP switches, from 200 up to 51,200 steps/rev. [3DM580S 3-Phase Digital Stepper Drive User Manual I User Manual 3DM580S +5](#)

CWD3M683 driver for the 57BHP79-Y580A-32 motor

Also 3-phase, with a similar opto-isolated PUL+/PUL-, DIR+/DIR- style input. Specs: input voltage 24–60 VDC, output current up to 8.3A, pulse frequency 0-200kHz, logic signal current 7-16mA typical 10mA. The vendor is explicit that PULSE and DIR signals should be sent as 5V, and if the signal source outputs more than 5V, a resistor must be added in the signal line, for example a 1.8kΩ resistor for a 24V output. Microstepping is set via DIP switches SW5-SW8, giving step/rev options from 400 up to 40,000. [Digital Stepper Drive CWD3M683 +3](#)

How this maps onto your wiring

- Signal type match: All three are compatible, the RDC6445S's OC pulse/dir outputs pair naturally with either driver's opto-isolated inputs. Wire it common-anode: tie PUL+/DIR+ on the driver to the controller's 5V supply pin, and PUL-/DIR- to the controller's OC pulse/dir output pins.
- Voltage: Straightforward here since the controller's pulse/dir outputs and both drivers' logic inputs are all designed around 5V, no resistor should be needed unless you deviate from the standard 5V reference.
- Grounding: Because both drivers use optical isolation on the control side, the control signal "ground" (the controller's 5V/common) doesn't need to be tied to the driver's motor-power ground. Just make sure the panel/chassis is earthed as CWD3M683's documentation stresses, and keep motor power wiring separate from signal wiring to avoid noise.

- Pulse edge/polarity: Confirm in the RDC6445S software whether pulses are set to rising or falling edge active, and match that against how you've wired PUL+/PUL- polarity on the driver.
- Microstepping vs. pulses/unit: Whatever steps/rev you set on the driver's DIP switches (e.g., 400, 1600, 3200... on the CWD3M683, or up to 51,200 on the 3DM580) must exactly match the pulses-per-unit value you configure in the RDC6445S's axis parameters, or your commanded distances will be off.

Signal path to understand: RD controller -> STEP/DIR command signals -> motor driver -> controlled phase current -> motor rotation. If feedback is used, it returns through the drive or controller according to the selected architecture.

C4. VFD / Inverter for a Spindle

- What is a variable-frequency drive (VFD), often called an inverter in CNC workshops?
- Why is a VFD used with many AC spindle motors? Address speed control, acceleration and deceleration, direction, and protection.
- Why is a VFD generally unnecessary for a pen-only plotter?

Your answer / notes:

What is a VFD?

A VFD (variable-frequency drive), often called an "inverter" in workshops, is a small electronic box that sits between the wall power and the spindle motor. Normal AC power comes in at a fixed frequency (like 50 or 60 Hz) and fixed voltage. The VFD converts this into AC power where it can freely control both the frequency and the voltage going out to the motor. Since a motor's speed depends on that frequency, changing the frequency changes the motor's speed.

Why spindles need a VFD:

- **Speed control** – Without a VFD, the motor would only run at one fixed speed (set by the wall power's frequency). The VFD lets you dial in almost any RPM, so one spindle motor can handle slow, tough cuts and fast, light cuts just by turning a dial or sending a command.
- **Smooth acceleration and deceleration** – If you switched a spindle motor straight on at full power, it would draw a huge current surge and jolt the mechanics. The VFD ramps the frequency up and down gradually, so the motor spins up and slows down smoothly, easier on the motor, electrical system.
- **Direction control** – To reverse an AC motor, you need to swap two of its phase wires. The VFD does this internally with a command, so you can reverse the spindle direction electronically.
- **Protection** – VFDs constantly monitor the current going to the motor. If something draws too much current (a stall, overload, or short), the VFD trips and cuts power before the motor overheats or burns out. This is protection you wouldn't have with a motor wired directly to the wall.

Why a pen plotter doesn't need one:

A pen plotter has no spindle, there's no rotating cutting tool at all. The "tool" is just a pen that moves up and down. The plotter's only moving parts are:

- Two stepper motors (X and Y axes), which are DC-driven and controlled by simple STEP/DIR signals, not AC power
- A small low-voltage servo that lifts and lowers the pen

None of these are AC motors running off mains power, so there's nothing for a VFD to control. A VFD's whole job, converting and varying AC power for an AC motor which simply doesn't apply here.

5. Final Challenge - Propose a Two-Axis Pen Plotter

Design a simple, safe, two-axis (XY) CNC pen plotter using the two components already available:

- One RD controller.
- Two stepper motors: one for the X axis and one for the Y axis.

Design scope: The machine will draw with a pen and will not cut material. A powered spindle and VFD are therefore outside the scope of the base design. If you include a pen-lift mechanism, explain how it will be actuated and controlled.

D1. Required Remaining Components

List all additional components required to complete the machine. For each item, state the quantity, a basic specification or assumption, and the reason it is needed.

Category	Component	Qty.	Key specification / assumption	Why it is required
Control / electrical	RDC6445S(EC) controller (already available)	1	CNC motion controller with dedicated X/Y STEP/DIR outputs and limit inputs	Generates and coordinates the X- and Y-axis motion commands for the plotter.
Power input / safety	AC power plug / mains input	1	AC input with L, N and PE (protective earth)	Provides mains power to the machine and the protective-earth connection.
Power input / protection	Circuit breaker (MCB)	1	Appropriately rated for the machine's AC input and wiring	Provides overcurrent and short-circuit protection for the mains-side wiring and equipment.
Safety / protection	Emergency-stop pushbutton	1	Latching mushroom type, NC contact; used with an appropriate enable/power-isolation circuit	Allows the operator to stop machine motion quickly in an emergency.
Control / electrical	Controller DC power supply	1	Regulated DC supply compatible with the RDC6445S(EC) requirements	Provides the required low-voltage DC power for the RD controller.
Control / electrical	24 V motor power supply	1	Regulated switching supply, approximately 24 V, 15 A, with suitable design margin	Supplies DC power to the two stepper drivers and motion system.
Control / electrical	3-phase stepper drivers	2	3DM580: 20-50 VDC; or CWD3M683: 24-60 VDC; current setting matched to the connected motor	Converts the RD controller's STEP/DIR commands into controlled 3-phase motor current.
Motion / electrical	Stepper motors (already available)	2	573S15-L for one axis and 57BHP79-Y580A-32 for the other; driver current limit set according to each motor's rated phase current	Provide rotary motion for the X and Y axes.
Safety / sensing	Home microswitches	2	One per axis; compatible with the RD controller's dedicated X/Y limit inputs	Establish repeatable X and Y home positions; software travel limits restrict normal travel.
Safety / grounding	Protective-earth wiring / bonding	1 set	PE from AC input bonded to PSU FG/earth terminals and applicable exposed metal frame/enclosure	Provides protective grounding for exposed conductive parts.
Mechanical frame	Gantry frame (assumed pre-built)	1 set	Rigid structure sized for the intended X/Y travel	Supports the motors, rails, belts, pulleys and pen mechanism.
Linear guidance	Linear rail + carriage block (or rod + bearing)	2 sets (X & Y)	e.g. MGN12 rail or suitable 8-12 mm rod/bearing arrangement	Guides each axis along a straight path and supports the mechanical load.
Motion transmission	GT2 timing belt	2	2 mm pitch; one belt for each axis	Transfers movement from each drive pulley to its corresponding carriage.
Motion transmission	20-tooth GT2 drive pulley	2	One mounted on each stepper-motor shaft	Converts motor rotation into belt movement.
Motion transmission	GT2 idler / tensioning pulley	2	One per axis; adjustable where practical	Provides belt return and allows correct belt tension.
Pen tool / lift	Pen holder assembly	1	Lightweight rigid holder suitable for the selected pen	Holds the pen securely and maintains its position during drawing.
Pen tool / lift	Pen-lift servo / actuator	1	Low-voltage servo with sufficient torque and travel for the pen mechanism	Raises the pen during travel moves and lowers it for drawing.
Control / electrical	Servo power supply / DC-DC converter	1*	Regulated output matched to the selected servo, if no suitable	Supplies the servo at its required voltage instead of applying the 24 V motor supply directly.

Category	Component	Qty.	Key specification / assumption	Why it is required
			regulated servo supply is already available	
Wiring / assembly	Terminal blocks, connectors and wiring	1 set	Appropriately rated mains, DC, motor and signal wiring	Provides secure electrical interconnections between the controller, drivers, motors, switches and supplies.

D2. System Architecture

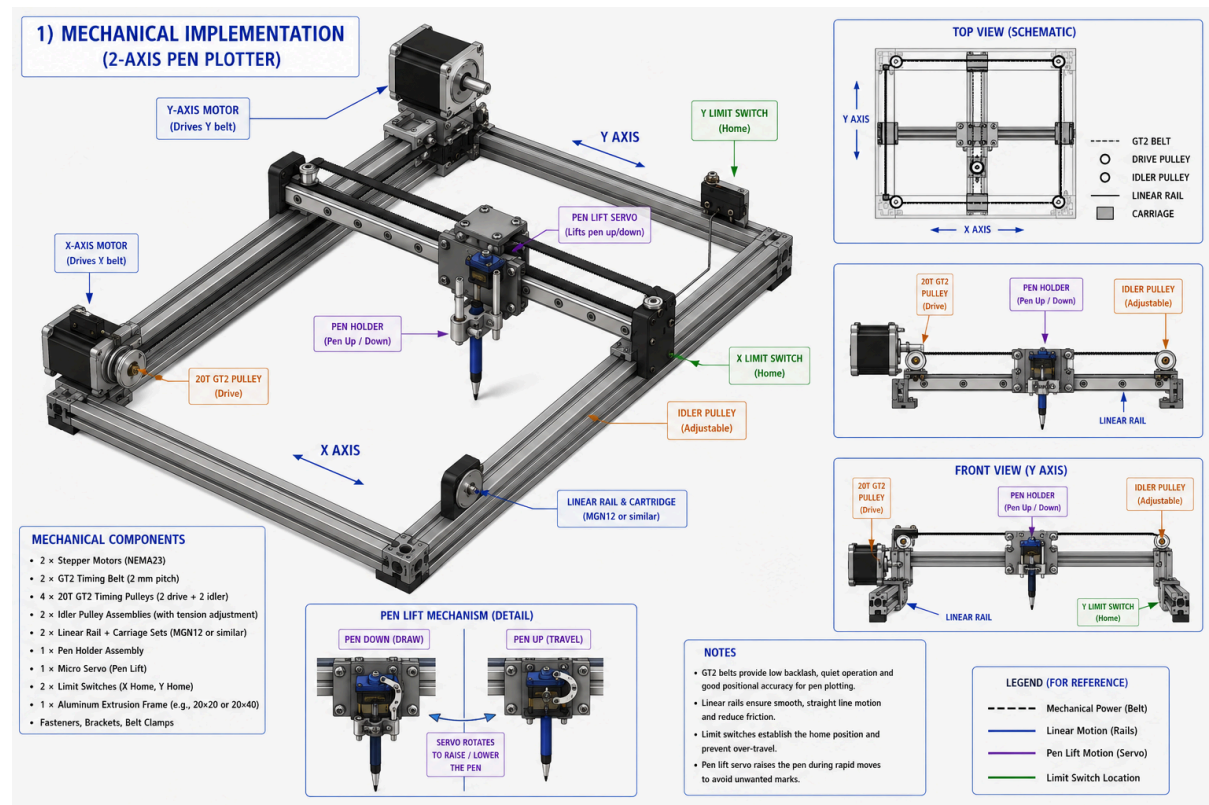
Create a system architecture sketch that shows the complete signal and power flow. Include:

- AC power input, isolation, and protection, where applicable.
- DC power supply or supplies, including assumed voltage and current ratings.
- Connections from the RD controller to two compatible stepper drivers using STEP/DIR signals.
- Motor-phase wiring from each driver to the X- and Y-axis motors.
- Home and limit switches, an emergency-stop circuit, grounding, and cable management.
- The selected linear guides and motion-transmission mechanisms for both axes.
- The pen holder and optional pen-lift mechanism.

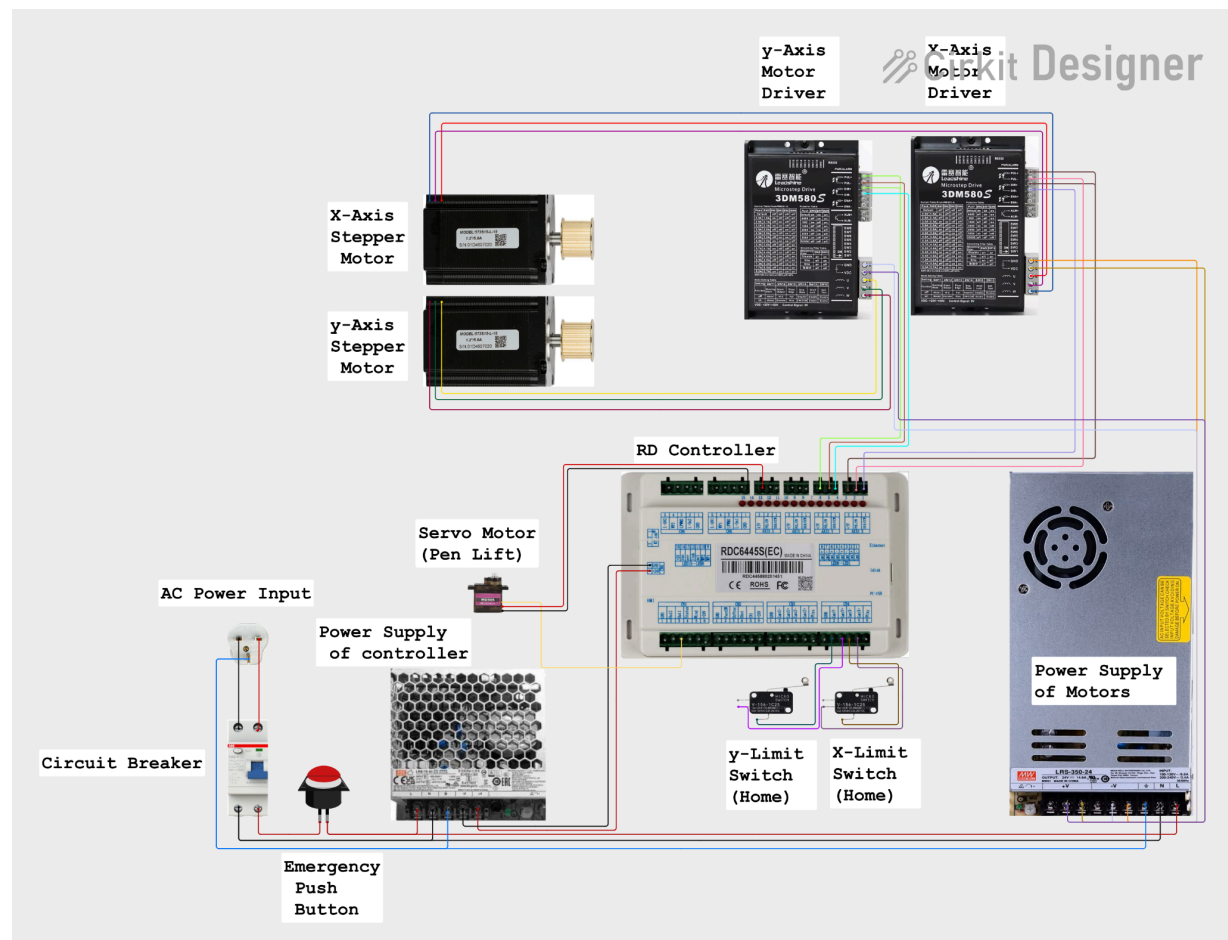
D2. System Architecture Sketch

Use the entire A4 page for your sketch. Show the power path, control-signal path, X and Y axes, motion-transmission mechanisms, safety devices, and optional pen-lift mechanism. Clearly label all major components and connections.

1- Mechanical Implementation:



2- Electrical Wiring:



6. Design Rationale and Submission Check

D3. Explain Your Choices

Question A: Why is your selected motion-transmission method appropriate for a pen plotter?

Response: Timing belt/pulley — pen-drag forces are small enough that belt compliance and backlash don't show up as visible error, and the belt's low mass/inertia lets the stepper accelerate quickly through curves for smooth strokes (full reasoning in A3).

Question B: How did you match the drivers and power supply to the stepper motors?

Response: Read each motor's rated phase current off its nameplate and set the matching driver's current-limit trimpot at or slightly below that value (roughly 80-90%) to control heating.

Chose driver supply voltage (24 V) well above the motor's rated voltage but within the driver's rating, since higher bus voltage pushes current into the winding faster and preserves torque at higher step rates.

Sized the PSU current as the sum of both drivers' set currents plus about 30% margin for the pen-lift servo and controller logic.

Question C: How will the machine establish its home position and prevent unsafe overtravel?

Response: On power-up/reset, the RD controller drives each axis slowly toward its home switch until it trips, backs off a small distance, and zeros its internal step count there. This establishes the machine coordinate origin. Unsafe overtravel during normal operation is prevented using software travel limits configured for the maximum permitted X- and Y-axis travel.

Question D: What resolution do you expect? Show how motor steps, microstepping, and transmission produce steps per millimetre.

Response: Steps per mm = (motor full steps/rev x microstepping) / (belt pitch x pulley tooth count).

Example: 1.8-degree stepper = 200 full steps/rev; driver set to 1/16 microstepping -> 3200 microsteps/rev; GT2 belt, 2 mm pitch, 20-tooth pulley -> 40 mm of travel per revolution.

Steps/mm = 3200 / 40 = 80 steps/mm, i.e. a theoretical step size of about 0.0125 mm (12.5 microns). Real repeatability will be coarser than this once belt compliance and backlash are accounted for.

Question E: What is the main technical risk in your design, and how would you test it?

Response: Main risk: silent lost/missed steps from the open-loop stepper system — caused by a snagged pen-lift servo, incorrect belt tension, or acceleration set too aggressively for the motor's available torque. Because there is no encoder feedback, the controller cannot detect this, so the drawing simply comes out shifted partway through.

Test: run a repeated square or diamond pattern from the same origin many times in a row and check whether the shape still closes on itself. Any progressive offset points to lost steps, and lets acceleration and driver current be tuned down before running a real job.

Submission Checklist

[Done] All four axis-motion methods researched and compared.

[Done] Motor-family scan completed.

[Done] Stepper vs servo comparison completed.

[Done] Open-loop, closed-loop, and encoder roles explained.

[Done] Incremental vs absolute encoders compared.

[Done] Driver purpose and STEP/DIR concept explained.

[Done] VFD purpose for spindle control explained.

[Done] Pen-plotter component list, system architecture, and design rationale completed.

Sources Consulted

1. CNCCookbook — "CNC Rack and Pinion vs Belt Drive or Ballscrew" — <https://www.cnccookbook.com/cnc-rack-and-pinion-vs-belt-drive-or-ballscrew/>
2. Design News — "Stepper vs Servo Motors: Pros, Cons, How to Choose the Right One" — <https://www.designnews.com/motors-actuators-conveyors/stepper-vs-servo-motors-pros-cons-how-to-choose-the-right-one>
3. RealPars — "Absolute vs Incremental Encoders: Key Differences Explained" — <https://www.realpars.com/blog/absolute-vs-incremental-encoder>
4. Techno CNC — "Ball Screws vs. Helical Rack and Pinion" — <https://www.technocnc.com/ball-screws-vs-rack-n-pinion/>
5. Ruida Technology — "RDC6445G/S Control System Manual" — <https://www.ruidacontroller.com/wp-content/uploads/2021/10/Users-Manual-of-RDC6445G-S-Control-System-V1.5.pdf>
6. Leadshine — "3DM580 Fully Digital Three Phase Stepping Driver User's Manual" — http://leadshineusa.com/UploadFile/Down/3DM580m_v1.0.pdf
7. Changzhou Chuangwei Motor & Electric — "CWD3M683 Digital Stepper Drive Datasheet" — <https://www.circuitspecialists.com/content/125531/CWD3M683.pdf>

Come prepared for the training session: Bring your completed research, source links, and any questions you would like to discuss. The CNC pen plotter's mechanical structure will already be assembled. Starting with this ready motion system, you will connect and wire the electrical control and power circuits, develop a practical solution for controlling the pen mechanism, and learn how to create and run drawings using the RD controller. You will then test the machine, evaluate its motion, and improve its performance by calculating and adjusting the pulse equivalent and other motion-control parameters.